Yuchen Bi

Curriculum Vitae

Education

2025–now **Postdoctoral Researcher (Advisor: Ernst Kuwert)**, *University of Freiburg*, Germany

2023–2025 **Postdoctoral Researcher (Advisor: Gang Tian)**, *Peking University (BICMR)*, China

2015–2023 **Ph.D. in Mathematics (Advisor: Jiayu Li)**, *University of Chinese Academy of Sciences (AMSS, CAS)*, China

2011–2015 **B.Sc. in Mathematics**, *Wuhan University*, China

Research Interests

Geometric analysis

Publications

Published

[1] Yuchen Bi, Jiayu Li, "The Prescribed Q-Curvature Flow for Arbitrary Even Dimension in a Critical Case", *Advances in Mathematics* 438 (2024), 109478.

[2] Yuchen Bi, Jiayu Li, Lei Liu, Shuangjie Peng, "The C^0-convergence at the Neumann boundary for Liouville equations", *Calculus of Variations and Partial Differential Equations* 62 (2023), no. 3, Paper No. 107.

[3] Yuchen Bi, Jie Zhou, "Optimal rigidity estimates for varifolds almost minimizing the Willmore energy", *Transactions of the American Mathematical Society* 378 (2025), no. 2, 943–965.

Preprints

[4] Yuchen Bi, Jintian Zhu, "Curvature-free Effects from Volume Growth and Ends-counting and Their Applications", arXiv:2605.12403.

[5] Yuchen Bi, Jintian Zhu, "Riemannian Penrose Inequality in All Dimensions", arXiv:2605.00680.

[6] Yuchen Bi, Tianze Hao, Shihang He, Yuguang Shi, Jintian Zhu, "A Proof for the Riemannian Positive Mass Theorem up to Dimension 19", arXiv:2603.02769.

[7] Yuchen Bi, Jie Zhou, "Linear Quantitative Rigidity for Almost-CMC Surfaces", arXiv:2601.09457.

[8] Yuchen Bi, Jie Zhou, "Quantitative Stability of the Clifford Torus as a Willmore Minimizer", arXiv:2511.19681.

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[9] Yuchen Bi, Jintian Zhu, "Mass-Capacity Inequality Modeled on Conformally Flat Manifolds", arXiv:2511.13215.

[10] Yuchen Bi, Jie Zhou, "Bi-Lipschitz Regularity of 2-Varifolds with the Critical Allard Condition", arXiv:2212.03043.

[11] Yuchen Bi, Jie Zhou, "Bi-Lipschitz Rigidity for L^2-Almost CMC Surfaces", arXiv:2212.02946.

Teaching Experience

2026 **Teaching Assistant**, *University of Freiburg*, Germany
Curves and Surfaces, Summer Semester 2026, with Ernst Kuwert.

2025–2026 **Lecture and Exercise Sessions**, *University of Freiburg*, Germany
Graduate course "Differential Geometry", Winter Semester 2025/26.